

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

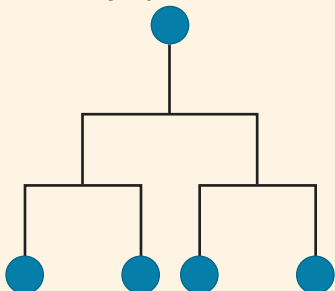
Levels 1-2: Remembering/Understanding

- The cortical reaction of sea urchin eggs functions directly in
 - the formation of a fertilization envelope.
 - the production of a fast block to polyspermy.
 - the generation of an electrical impulse by the egg.
 - the fusion of egg and sperm nuclei.
- Which of the following is common to the development of both birds and mammals?
 - holoblastic cleavage
 - epiblast and hypoblast
 - trophoblast
 - gray crescent
- The archenteron develops into
 - the mesoderm.
 - the endoderm.
 - the placenta.
 - the lumen of the digestive tract.
- What structural adaptation in chickens allows them to lay their eggs in arid environments rather than in water?
 - extraembryonic membranes
 - yolk
 - cleavage
 - gastrulation

Levels 3-4: Applying/Analyzing

- If an egg cell were treated with EDTA, a chemical that binds calcium and magnesium ions,
 - the acrosomal reaction would be blocked.
 - the fusion of sperm and egg nuclei would be blocked.
 - the fast block to polyspermy would not occur.
 - the fertilization envelope would not form.
- In humans, identical twins are possible because
 - extraembryonic cells interact with the zygote nucleus.
 - convergent extension occurs.
 - early blastomeres can form a complete embryo if isolated.
 - the gray crescent divides the dorsal-ventral axis into new cells.
- Cells transplanted from the neural tube of a frog embryo to the ventral part of another embryo develop into nervous system tissues. This result indicates that the transplanted cells were

(A) totipotent.	(C) differentiated.
(B) determined.	(D) mesenchymal.
- DRAW IT** Each blue circle in the figure below represents a cell in a cell lineage. Draw two modified versions of the cell lineage so that each version produces three cells. Use apoptosis in one of the versions, marking any dead cells with an X.



Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Evolution in insects and vertebrates has involved the repeated duplication of body segments, followed by fusion of some segments and specialization of their structure and function. In vertebrates, what anatomical features reflect segmentation?
- SCIENTIFIC INQUIRY** The “snout” region of a salamander has a mustache-shaped structure called a balancer, whereas that of a frog tadpole does not. When you transplant tissue from the side of a young salamander embryo to the snout of a frog embryo, the tadpole that develops has a balancer. If you use a slightly older salamander embryo as the donor, no balancer forms. Propose a hypothesis to explain these results and explain how you might test your hypothesis.
- SCIENCE, TECHNOLOGY, AND SOCIETY** Scientists can now make identical copies, or clones, of animals ranging from dairy cows to pet cats. Propose a few arguments for and against this application of discoveries about embryonic development.
- WRITE ABOUT A THEME: ORGANIZATION** In a short essay (100–150 words), describe how the emergent properties of the cells of the gastrula direct embryonic development.
- SYNTHESIZE YOUR KNOWLEDGE**



Occasionally, two-headed animals such as this turtle are born. Thinking about the occurrence of identical twins and the property of totipotency, explain how this might happen.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom

What is the molecular basis for limb regrowth in vertebrates?

Go to “Limb Regeneration: Fact or (Science) Fiction?”
at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.